

**DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY,
LONERE**

Regular & Supplementary Winter Examination-2023

Course: B. Tech. Class: Final Year Semester : VII

Branch : Electrical Engineering / Electrical Engineering (Electronics and Power)/ electrical & Electronics Engg / Electrical & Power Engineering

Subject Code & Name: Electric and Hybrid Electric Vehicles[BTEEOE705B]

Max Marks: 60 Date: 11 / 01 /2024 Duration: 3 Hr.

Instructions to the Students:

- 1. All the questions are compulsory.*
- 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.*
- 3. Use of non-programmable scientific calculators is allowed.*
- 4. Assume suitable data wherever necessary and mention it clearly.*

(Level/CO) Marks

Q.1 Solve Any Two of the following.

12

- | | | | |
|----|--|------------|---|
| A) | Explain about the history of hybrid and electric vehicles? | Remember | 6 |
| B) | Explain about the social and environmental importance of EV and HEV? | Understand | 6 |
| C) | Explain about the vehicle transmission characteristics? | Understand | 6 |

Q.2 Solve Any Two of the following.

12

- | | | | |
|----|---|------------|---|
| A) | How the hybrid electric vehicles are classified based on general definition? | Understand | 6 |
| B) | Explain the different power flow control modes of a typical parallel hybrid system with the help of block diagrams? | Applying | 6 |
| C) | Explain the fuel efficiency analysis of hybrid drive trains and electric drive trains? | Understand | 6 |

Q.3 Solve Any Two of the following.

12

- | | | | |
|----|---|----------|---|
| A) | With the help of block diagram explain the major components of an electric vehicle | Remember | 6 |
| B) | With the help of neat diagram explain the four-quadrant chopper-based speed control of DC motors? | Applying | 6 |
| C) | Discuss the possible power converter topologies that can be used in Induction motor drive for EV/HEV? | Applying | 6 |

- Q.4 Solve Any Two of the following.** **12**
- A) What are factors affecting the performance of batteries used in EVs? Understand **6**
 - B) How is it possible to use fly wheel as an energy storage device for electric vehicle? Understand **6**
 - C) Give the advantages and disadvantages of fuel cells? What is meant by c - rating of a battery? If an 100Ah battery is rated c5, what would be its discharge current expressed as 0.5Cs? Applying **6**
- Q.5 Solve Any Two of the following.** **12**
- A) Explain briefly the electrical and mechanical constraints to be considered while sizing an electrical machine for an EV? Understand **6**
 - B) What are the points to be considered for choosing the energy storage technology for hybrid electric vehicle? Understand **6**
 - C) Draw the block diagram of a general Fuzzy Logic Controller (FLC) and show the core components of the FLC and the inputs and outputs relevant to a hybrid electric vehicle control. Applying **6**

***** End *****